

# Yiren(Spike) Ding

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## WORK EXPERIENCE

### Senior Software Engineer

PayPal. San Jose CA

Jul.2024 – Current

- Designed and implemented a multi-level real-time anomaly detection framework for large-volume transaction data, processing over a million records daily.
- Leveraged advanced statistical method and machine learning models to forecast thousands of dimensions of data, and identify unusual patterns.
- Designed and implemented interactive data visualization dashboards for monitoring transactions, anomalies and reconciliation results, using both D3.js and Tableau.
- Built MCP Servers and Agents for anomaly exploration, source tracing, and clustering using FastMCP and CrewAI

### Software Engineer / AI Research Intern

PayPal. Remote /San Jose, CA

May. 2022/2023 – Aug. 2022/2023

- Designed and implemented an AI chatbot for PayPal's internal app deployment platform. The chatbot was built using TensorFlow, NLP.js, and an independent search engine.
- Developed an interactive visualization dashboard for querying payment transactions, integrated into the transaction explorer tool within the payment tools framework, utilizing D3 and React.
- Built an anomaly detection algorithm and a real-time reporting system to monitor transaction latency.

### Research Assistant

VIEW Lab, Worcester Polytechnic Institute. Worcester, MA

Aug. 2019 – Jan. 2025

- Architected and developed a scalable platform for conducting empirical studies on interactive data visualizations, using React, Firebase, and D3. The platform consists of a domain specific language (DSL), a compiler for study flow, built-in component library, data analytics platform and automated participants activity logging.
- Designed and built innovative data visualizations via D3.js in various domains. i.e., AI-assist interactive classroom observation system, racing bar chart for COVID cases, cartograms for spatial-temporal datasets.
- Conducted crowdsourcing experiments to model user perception of visualization channels. Applied multi-level Bayesian modeling to capture individual differences and support personalized interfaces.
- Pioneered the use of large language models (LLMs) to generate structured visualizations and assessments for evaluating data visualization literacy.

## EDUCATION

Ph.D. in Computer Science

2019-2025

Worcester Polytechnic Institute, Worcester MA

Master of Science in Computer Science

2017-2019

Northeastern University, Boston MA

## SKILLS

**Programming Languages:** JavaScript, TypeScript, Python, Java, C#, R, Go

**Frameworks & Libraries:** React, D3.js, Plotly, Mantine, Spring Boot, BQML

**Cloud Services & Database:** AWS, Google Cloud, Firebase, MongoDB, PostgreSQL, MySQL, BigQuery

**AI & ML:** PyTorch, LLM Fine Tuning, MCP, Agent Orchestration